T7 - Indian Wildlife Identifier: Venomous vs. Non-Venomous Snakes

Statistical Methods in AI
Assignment-3 Report

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1 Introduction

1.1 Context and Motivation

India accounts for the highest global burden of snakebites. According to the World Health Organization (WHO), there are an estimated 58,000 snakebite deaths annually in India out of approximately 3 to 4 million envenoming cases. The vast majority of these fatal and medically significant bites are attributed to the "Big Four" venomous snake species: the Indian Spectacled Cobra (*Naja naja*), Common Krait (*Bungarus caeruleus*), Russell's Viper (*Daboia russelii*), and Saw-scaled Viper (*Echis carinatus*).

Rapid and accurate identification of a snake can provide critical ecological awareness for hikers, trekkers, and rural residents. To address this, we developed a web-based wildlife identification system capable of classifying images of Indian snakes as venomous or non-venomous in real-time.

1.2 Project Scope

The project delivers a full-stack web application built with a **React** frontend and a **FastAPI** backend. The core classification engine is powered by an `efficientnet_b0` convolutional neural network, fine-tuned specifically on Indian snake species.

To augment the user experience with educational value, the system outputs the top-3 species predictions alongside a binary (venomous/non-venomous) classification. Furthermore, the pipeline integrates with the Gemini API to fetch an informative paragraph and the IUCN conservation status for the identified snake. To optimize performance and reduce API latency, these facts are queried once and cached locally as JSON.

Medical and Safety Disclaimer:
*This artificial intelligence system is designed strictly for educational, project, and ecological awareness purposes. It is **not** a substitute for professional herpetological identification or emergency medical advice. In the event of a snakebite, users must seek immediate professional medical attention at the nearest healthcare facility, regardless of the model's prediction.*

2 Data

2.1 Dataset Overview

The primary data source for this project is the **"Snake Dataset - India"** (sourced via Kaggle). The dataset is composed of images of various snake species indigenous to the Indian subcontinent, encompassing highly venomous species (such as cobras and vipers) as well as non-venomous species (such as rat snakes and green tree vines).

The raw dataset contains a total of **2,044 images**, with each image originally possessing a uniform spatial resolution of 400×400 pixels.

2.2 Data Distribution & Class Imbalance

In a binary classification problem, analyzing the class distribution is critical to prevent the model from developing a biased prior probability toward the majority class.

The original dataset has **1201** images of venomous snakes and **843** images of non-venomous snakes.

2.3 Preprocessing and Splits

To prepare the data for the `EfficientNet_B0` architecture, uniform preprocessing was applied. The images were resized to the standard input dimension expected by the base model to ensure consistent tensor shapes during batch processing.

To rigorously evaluate the model during training, the dataset was restructured into an approximate **70-15-15** split for Training, Validation and Testing.

3 Methodology

3.1 Model Architecture

The core classification engine is built upon the EfficientNet-B0 architecture, chosen for its optimal balance between high accuracy and low computational resource requirements. The model was initialized with pre-trained ImageNet weights (IMAGENET1K_V1). To adapt the network for our specific binary classification task, the final fully connected layer of the classifier was replaced with a new linear layer matching the number of target classes (venomous vs. non-venomous).

3.2 Data Augmentation and Preprocessing

To improve the model's ability to generalize to unseen real-world scenarios and prevent overfitting, dynamic data augmentation was applied exclusively to the training set. The augmentation pipeline included:

- Resizing images to 256x256 pixels followed by a random crop to 224×224 pixels.
- Random horizontal flipping and random rotations (up to 15 degrees).
- Color jittering (adjusting brightness, contrast, saturation, and hue) to simulate various lighting conditions in the wild.

The validation and test sets were strictly resized and center-cropped to 224×224 without augmentation to ensure objective evaluation. All images were normalized using standard ImageNet mean and standard deviation metrics.

3.3 Handling Class Imbalance

To mitigate the moderate class imbalance identified in the dataset (leaning towards venomous snakes), class weights were computed based dynamically on the training data distribution. These weights were passed into the CrossEntropyLoss function, heavily penalizing misclassifications of the minority class and preventing the model from becoming biased toward the majority class.

3.4 Training Strategy and Transfer Learning

The training process utilized a two-phase transfer learning strategy to preserve the robust feature-extraction capabilities of the pre-trained backbone:

- Phase 1 (Warm-up): For the first 3 epochs, the entire feature extractor backbone was frozen. Only the newly initialized classifier layer was trained using the Adam optimizer with a learning rate of 3×10^{-4}
- Phase 2 (Fine-tuning): The backbone was unfrozen, and the entire network was fine-tuned using a reduced learning rate of 1×10^{-4}

3.5 Optimization and Regularization

To optimize convergence, a **ReduceLROnPlateau** scheduler was implemented to decay the learning rate by a factor of 0.3 if the validation loss stagnated for 2 consecutive epochs. Furthermore, an Early Stopping mechanism was employed with a patience of 5 epochs. Instead of monitoring simple accuracy or loss, early stopping monitored the Validation

F1-score, ensuring the saved model maintained a robust balance between precision and recall.

4 Evaluation Metrics

The performance of the model was evaluated using multiple metrics to ensure a reliable and safety-aware assessment. In the context of snake identification, different types of errors carry significantly different real-world consequences, making it essential to look beyond overall accuracy.

The model is evaluated against the ground truth labels (venomous vs. non-venomous) using held-out validation and test data to assess generalization.

The primary concern is correctly identifying venomous snakes, making **recall for the venomous class the most important metric**, as false negatives (i.e., predicting a venomous snake as non-venomous) can have serious, potentially fatal consequences. Precision is also considered to avoid excessive false alarms, but it is treated as secondary to recall. The F1-score provides a balanced view of performance, while accuracy is used only as a coarse indicator due to class imbalance. Overall, the evaluation prioritizes minimizing dangerous misclassifications over maximizing overall correctness.

To formalize these metrics, we first define the components of the confusion matrix in the specific context of this study:

- **True Positive (TP):** Venomous snakes correctly identified as venomous.
- **True Negative (TN):** Non-venomous snakes correctly identified as non-venomous.
- **False Positive (FP):** Non-venomous snakes incorrectly identified as venomous (false alarm).
- **False Negative (FN):** Venomous snakes incorrectly identified as non-venomous (the critical error).

Accuracy measures the overall proportion of correct predictions across both classes. Due to the class imbalance, this is used only as a coarse indicator.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Recall (Sensitivity / True Positive Rate) is the most critical metric for this task. It measures the proportion of actual venomous snakes that were correctly identified, emphasizing the minimization of false negatives.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

Precision (Positive Predictive Value) measures the accuracy of the positive (venomous) predictions. It helps evaluate how many false alarms the model produces.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

F1-Score is the harmonic mean of precision and recall. It provides a single metric that balances both the need to catch venomous snakes (recall) and the desire to avoid excessive false alarms (precision).

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \tag{4}$$

5 Results

The model was rigorously evaluated on the unseen hold-out Test Set using the best weights saved during the training phase. Based on the performance trajectories and early-stopping criteria, the optimal model weights were successfully captured at Epoch 11, achieving the highest recall before any significant overfitting occurred.

The final evaluation on the test set yielded the following key metrics:

- Test Accuracy: 89.58% (318 correct predictions out of 355 total test samples)
- Test F1-Score: 0.9227

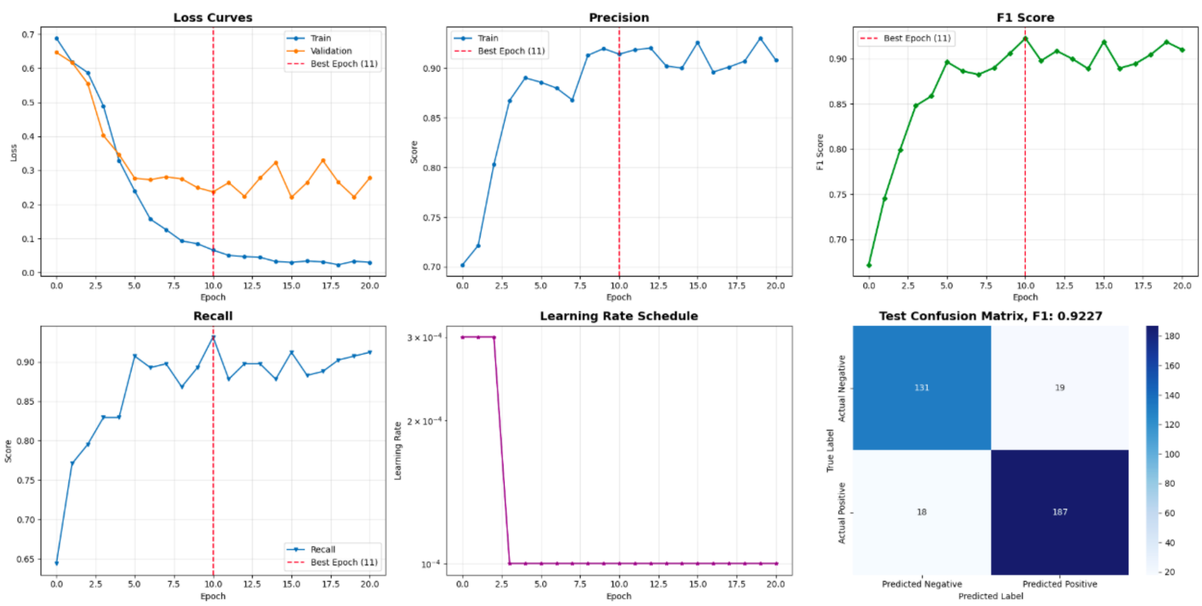

Figure 1: Training and validation metrics across epochs, alongside the final test set confusion matrix. The red dashed line indicates the optimal model checkpoint chosen by early stopping at Epoch 11.

5.1 Training and Validation curve analysis

The generated learning curves provide deep insights into the model's convergence behavior:

- Loss Curves: The training loss decreased steadily over the 20 epochs, indicating that the model was effectively learning feature representations. The validation loss tracked closely with the training loss initially and reached its absolute minimum at Epoch 11. Beyond Epoch 11, the validation loss began to exhibit slight oscillations and a mild upward trend, indicative of the onset of overfitting. The early-stopping mechanism correctly identified Epoch 11 as the optimal checkpoint, ensuring excellent generalization.

- Metric Curves (Precision, Recall, F1): The Validation Precision, Recall, and F1-score all showed rapid improvement during the first 5 epochs, stabilizing at high values. Peak performance across all three metrics perfectly aligned at Epoch 11, with the F1-score peaking at approximately 0.925 on the validation set.

- Learning Rate Schedule: The learning rate plot visually confirms the two-phase transfer learning strategy. For the first 3 epochs (the "warm-up" phase with a frozen backbone), the learning rate was maintained at 3×10^{-4}. At Epoch 4, upon unfreezing the backbone for fine-tuning, the learning rate appropriately stepped down to 1×10^{-4}, allowing the model to make finer adjustments to the pre-trained weights without aggressively distorting them.

5.2 Confusion Matrix and Safety Implications

The test-set Confusion Matrix provides a granular look at the classification distribution:

- True Positives (Actual Positive, Predicted Positive): 187
- True Negatives (Actual Negative, Predicted Negative): 131
- False Positives (Actual Negative, Predicted Positive): 19
- False Negatives (Actual Positive, Predicted Negative): 18

Note: In the context of this medical/safety-critical application, 'Positive' typically designates the dangerous (Venomous) class.

From a safety perspective, the most critical metric is the reduction of False Negatives (a venomous snake incorrectly classified as non-venomous). Out of the samples evaluated, only 18 instances were falsely predicted as negative, resulting in a highly favorable recall rate for the positive class (91.2%). While there were 19 False Positives (non-venomous snakes flagged as venomous), this is an acceptable trade-off; in a real-world ecological or hiking scenario, it is strictly preferable for the model to cautiously over-predict danger rather than fail to warn a user of a potentially lethal snake.

Overall, the F1-score of 0.9227 demonstrates that the EfficientNet-B0 model successfully balanced precision and recall, proving highly effective for this binary wildlife identification task.

6 Limitations

Despite the robust performance of the EfficientNet-B0 classifier, several limitations persist:

- Visual Mimicry (Batesian Mimicry): Several non-venomous snake species in the Indian subcontinent (e.g., the Common Wolf Snake) have evolved to mimic the banding patterns of highly venomous species (e.g., the Common Krait). The model may struggle to differentiate these without high-resolution analysis of microscopic scale structures (like the loreal shield), leading to false positives.
- Context and Out-of-Distribution Data: The model inherently assumes the presence of a snake in the image. If fed an image of an empty forest floor or a different reptile, it is forced to predict either venomous or non-venomous, lacking an "Unknown/No Snake" rejection class.
- Real-world Image Quality: The dataset primarily consists of well-lit, centered subjects. In real-world emergency scenarios, images are often blurry, taken in low light (many venomous snakes are nocturnal), or heavily occluded by foliage, which may degrade prediction confidence.

7 Future Work

While the current system demonstrates effective performance for binary classification of venomous and non-venomous snakes, several avenues exist for further improvement:

- **Species-level classification:** Extending the model to identify specific snake species rather than performing binary classification would significantly enhance its ecological and practical value.
- **Object detection or segmentation:** Integrating localization methods could allow the system to focus on the snake within complex scenes, improving classification accuracy.
- **Mobile and edge deployment:** Optimizing the model for lightweight inference would enable real-time usage in field conditions through mobile devices.
- **Uncertainty estimation:** Introducing confidence calibration or uncertainty-aware predictions would improve safety by flagging low-confidence outputs.

8 Conclusion

This project successfully developed an AI-driven, web-based tool for classifying venomous and non-venomous snakes native to the Indian subcontinent. By leveraging an EfficientNet-B0 architecture with a tailored transfer learning strategy, data augmentation, and class-weighted loss, the model achieved strong generalization capabilities. By prioritizing the F1-score and Recall during the early-stopping phase, the system aligns with critical medical and safety paradigms—ensuring that dangerous venomous snakes are accurately flagged.

Integrated with a FastAPI backend, a React frontend, and the Gemini API for fetching real-time ecological facts, the project transcends a standard machine-learning classification task to deliver a comprehensive, educational wildlife identifier. This lays a strong foundation for future mobile deployments aimed at aiding hikers, rural workers, and wildlife enthusiasts in real-time ecological awareness.

9 Screenshots

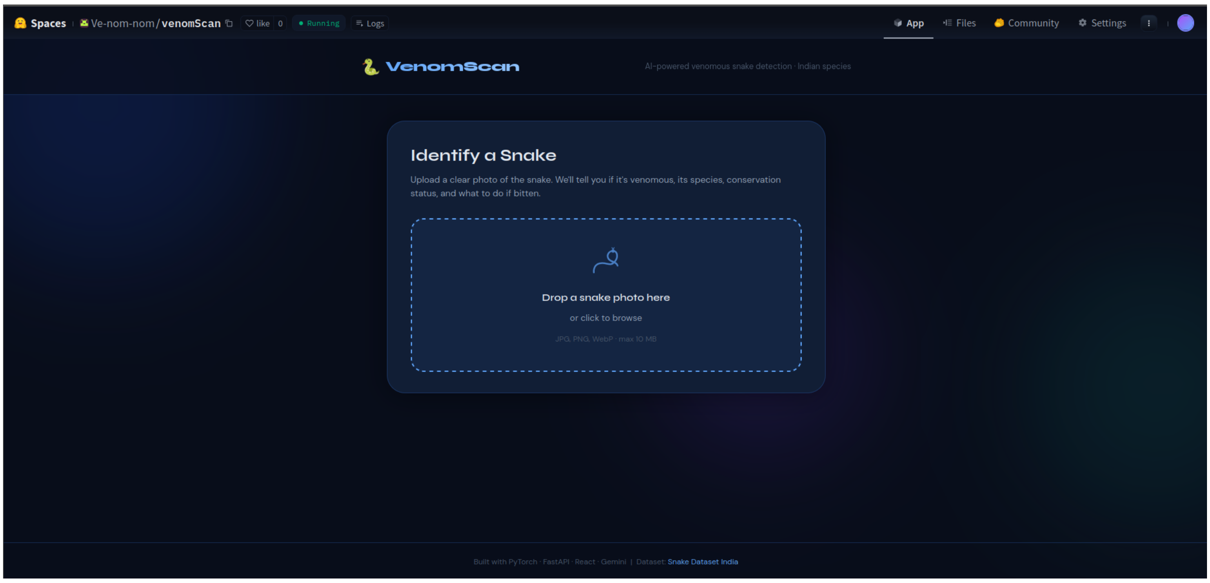

Figure 2: Main Dashboard

10 About and Links

10.1 Team Members

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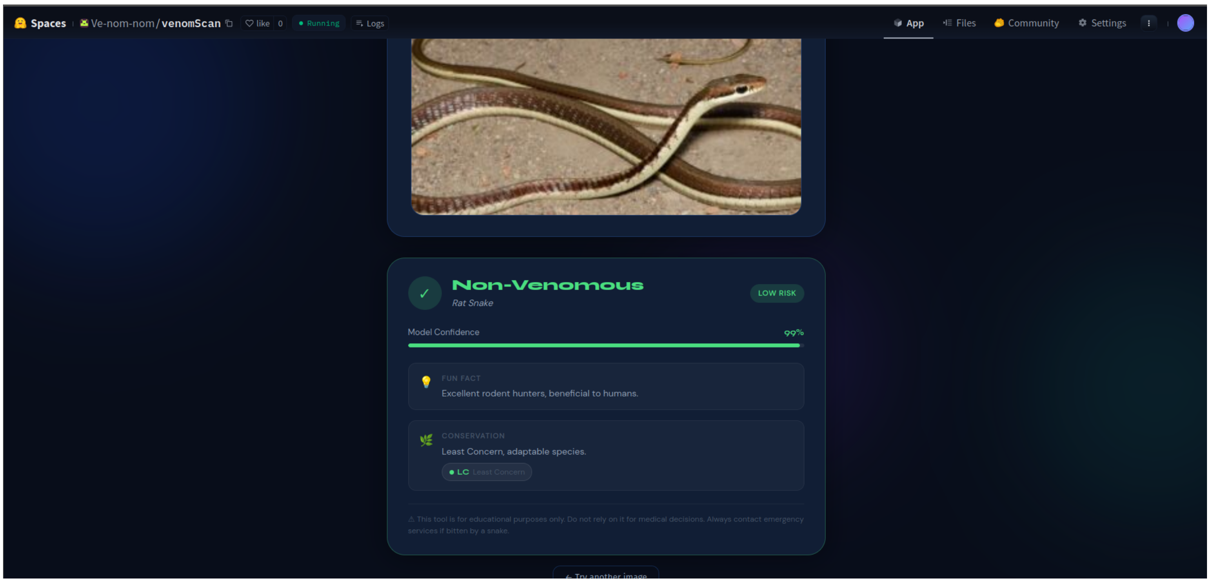

Figure 3: Non Venomous Snake case

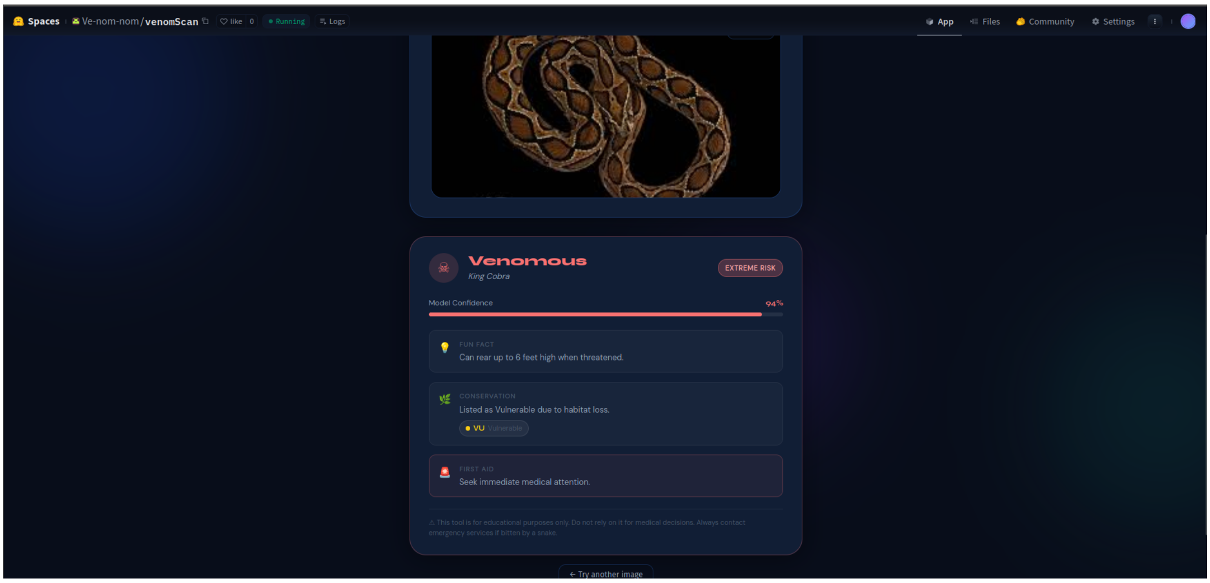

Figure 4: Venomous Snake case

10.2 Links

- Github: https://github.com/anushkasharma2005/Ve-nom-nom
- Live Website: https://huggingface.co/spaces/Ve-nom-nom/venomScan
- Trained Model: https://huggingface.co/Ve-nom-nom/venom_model

11 References

- Thakur, A., & Kaur, G. (2025). Detection of venomous and non-venomous snakes using convolutional neural network in deep learning. In Proceedings. https://doi.org/10.21467/proceedings.7.6.22